



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin

Ollscoil Átha Cliath | The University of Dublin

Faculty of Engineering, Mathematics and Science

School of Computer Science & Statistics

Integrated Computer Science
Integrated Engineering
Computer Science and Business
Computer Science and Language

Semester 1 2024

Year 4

11th December 2024

Computer Vision

17:00-19:00

Dr Kenneth Dawson-Howe

Instructions to Candidates:

Attempt two questions. All questions carry equal marks. Each question is scored out of a total of 50 marks.

For Application questions, a series of computer vision operations must be described to solve the application problem. You **MUST NOT** use neural networks/deep learning as part of your solution. The input to and output from each technique used must be clearly and precisely stated. How each technique is used within the context of the application must also be described including the setting of any parameters.

For Compare and Contrast type questions marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques.

For Explanation and Purpose type questions you are required to explain the terms provided including the context in which they are used and their purpose. Detailed descriptions are required.

In all questions you must describe computer vision theory and should not refer to code or library calls (OpenCV or any other library).

1. (a) **[APPLICATION QUESTION]** Describe how you would automatically locate and read license plates in images such as that shown below. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique. You must NOT use neural networks or deep learning as part of your solution.

[30 marks]



1. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:

- RANSAC to find lines.
- Hough transform for lines.
- Contour following followed by recursive boundary splitting.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[20 marks]

2. (a) **[APPLICATION QUESTION]** Describe how you would automatically locate and track the coloured helmets of horse riders in a video feed from a stationary camera which gives views like those below. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique.

[25 marks]



Photo by Mathew Schwartz released under the Unsplash License



Photo by Jeff Griffith released under the Unsplash License

2. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:

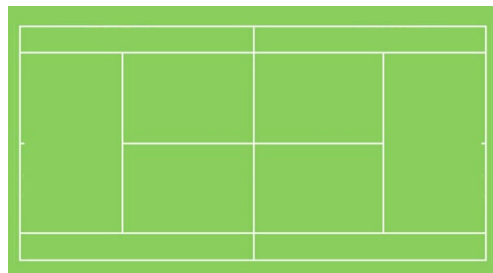
- Robust Object Recognition with a cascade of classifiers.
- Scale Invariant Feature Transform.
- Principal Components Analysis.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[25 marks]

3. (a) **[APPLICATION QUESTION]** Given an image of a tennis court (such as that shown below) describe how you would automatically locate the court (i.e. the white lines) using edge detection and transform it to a plan view. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique. You must NOT use neural networks or deep learning as part of your solution.

[25 marks]



PLAN VIEW



COMPLETE IMAGE OF THE COURT

3. (b) **[EXPLANATION & PURPOSE QUESTION]** Explain the following terms, the context in which they are used and their purpose:

- Hidden layers.
- Loss functions.
- Region Based CNNs (R-CNN).
- Recurrent Neural Networks (RNN).

For each term you must provide an explanation of the term (using diagrams if appropriate), the context within which it is used and also describe its purpose.

[25 marks]